Hall effect Anemometer

Laboratory of Sensors - Prof. Alexandro Catini

Nazanin Taherpoursomeh 1 August 2026

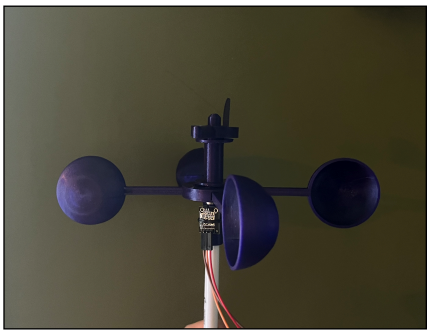


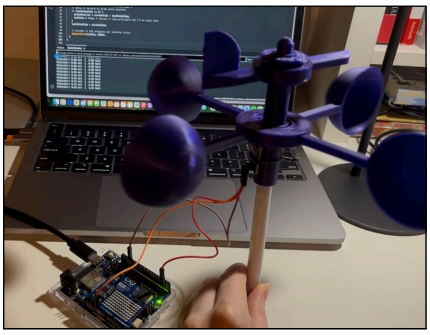
Introduction

For centuries, anemometers have been used to measure wind speed, velocity, and direction. The first mechanical design was introduced in 1450 by the Italian architect Leon Battista Alberti, who used a swinging disk to gauge wind force. Later, in 1846, John Thomas Romney Robinson developed the familiar four-cup rotating anemometer, which utilized aerodynamic drag force. Observers manually recorded the number of rotations over a set period to determine average wind speed.

In this project, we utilize the same underlying mechanical principle, integrated with physical computing to automate rotation counting. The physical assembly consists of four cups spaced at 90 degree angle intervals, a shaft, ball bearings for reduced friction, a protective enclosure, and a directional indicator. Wind kinetic energy is converted into electrical signals using a digital Hall effect sensor paired with an Arduino Uno R4 WiFi. Two Neodymium magnets are mounted on the rotating assembly 3mm from the sensor to ensure clear digital pulses. The sensor operates via an open-collector transistor stage with an integrated 10k ohms pull-up resistor to the 5V power rail; this ensures a default logic-HIGH state (5V) that drops cleanly to logic-LOW (0V) whenever a magnet passes the sensor face.

This report evaluates the performance of the developed anemometer under various ambient conditions, including light exposure variations, high humidity, elevated temperatures, and environmental resistance factors such as water and sand ingress. Additionally, the sensor analysis section details the calibration response curve across multiple testing intervals, the limit of detection, resolution and signal noise characteristics—providing insight into the operational reliability and performance boundaries of digital magnetic transducers.

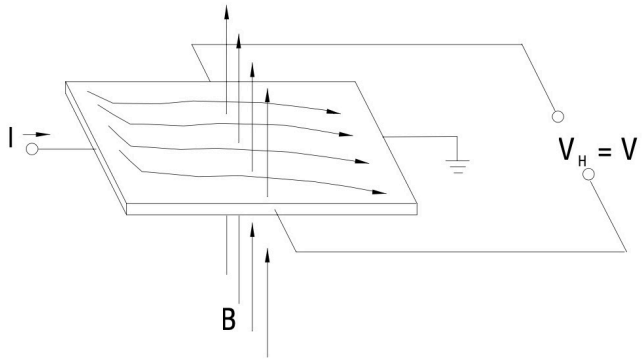




Theory of the Hall effect

When a current-carrying conductor is placed into a magnetic field, a voltage will be generated perpendicular to both the current and the field. This principle is known as the Hall effect.

When a perpendicular magnetic field is present, as shown in the image, Lorentz force is exerted on the current. This force disturbs the current distribution, resulting in a potential difference (voltage) across the output. This voltage is the Hall voltage (VH). The Hall voltage is proportional to the vector cross product of the current (I) and the magnetic field (B). It is on the order of 7 μv/Vs/gauss in silicon and thus requires amplification for practical applications.



$$V_H \propto \mathbf{I} \times \mathbf{B}$$

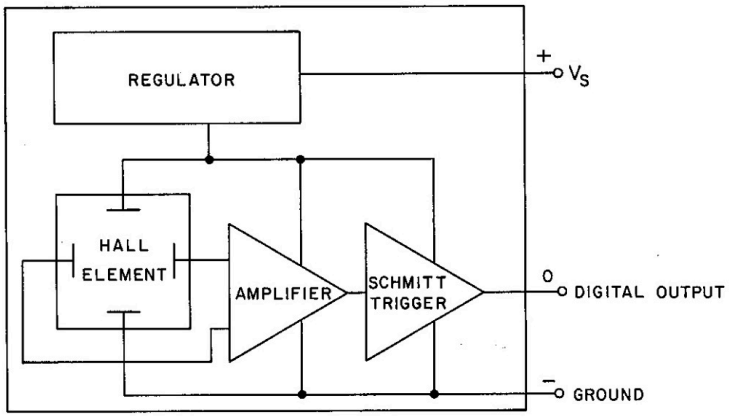
Basic Hall effect sensors

The Hall element is the basic magnetic field sensor. It requires signal conditioning to make the output usable for most applications. The signal conditioning electronics needed are an amplifier stage and temperature compensation. Voltage regulation is needed when operating from an unregulated supply. The image below illustrates a basic Hall effect sensor. If the Hall voltage is measured when no magnetic field is present, the output is zero. However, if voltage at each output terminal is measured with respect to ground, a non-zero voltage will appear. This is the common mode voltage (CMV), and is the same at each output terminal. It is the potential difference that is zero. The amplifier shown in the image must be a differential amplifier so as to amplify only the potential difference – the Hall voltage. The Hall voltage is a low-level signal on the order of 30 microvolts in the presence of a one gauss magnetic field. This low-level output requires an amplifier with low noise, high input impedance and moderate gain. A differential amplifier with these characteristics can be readily integrated

with the Hall element using standard bipolar transistor technology. Temperature compensation is also easily integrated.

Digital output sensors

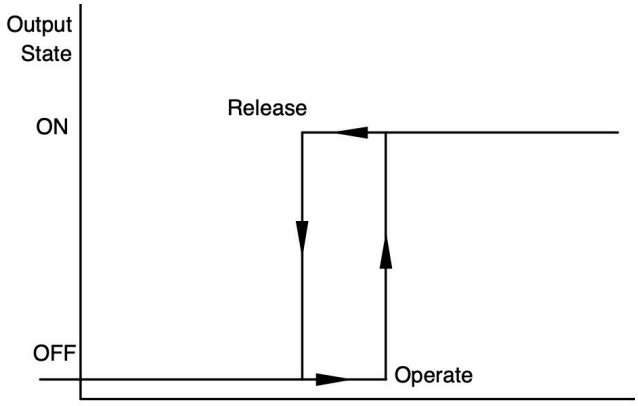
This sensor has an output that is just one of two states: ON or OFF. The basic analog output device illustrated in the image can be converted into a digital output sensor with the addition of a Schmitt trigger circuit. The picture illustrates a typical internally regulated digital output Hall effect sensor.



The Schmitt trigger compares the output of the differential amplifier with a preset reference. When the amplifier output exceeds the reference, the Schmitt trigger turns on. Conversely, when the output of the amplifier falls below the reference point, the output of the Schmitt trigger turns off.

Transfer function

The transfer function for a digital output Hall effect sensor incorporating hysteresis is shown in the figure. Hysteresis is included in the Schmitt trigger circuit for jitter-free switching. Hysteresis results from two distinct reference values which depend on whether the sensor is being turned ON or OFF.

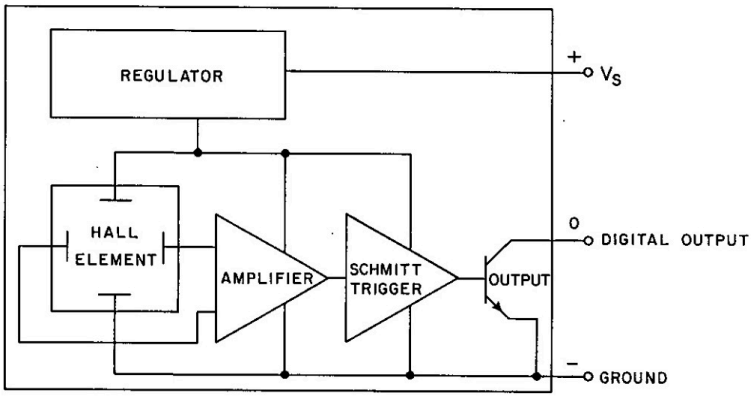


The principal input/output characteristics are the operate point, release point and the difference between the two or differential. As the magnetic field is increased, no change in the sensor output will occur until the operate point is reached. Once the operate point is reached, the sensor will change state. Further

increases in magnetic input beyond the operate point will have no effect. If magnetic field is decreased to below the operate point, the output will remain the same until the release point is reached. At this point, the sensor's output will return to its original state (OFF). The purpose of the differential between the operate and release point (hysteresis) is to eliminate false triggering which can be caused by minor variations in input.

NPN (Current sinking) Digital output sensor

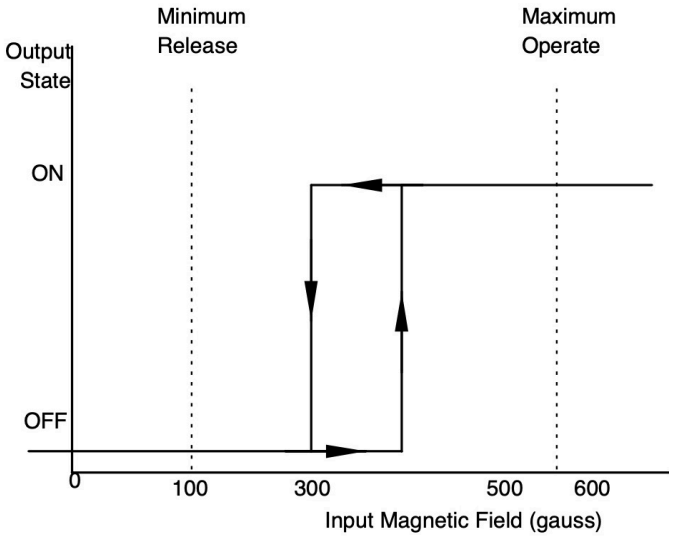
As with analog output Hall effect sensors, an output transistor is added to increase application flexibility. This output transistor is typically NPN (current sinking)



Power supply characteristics

Digital output sensors are available in two different power supply configurations - regulated and unregulated. Most digital Hall effect sensors are regulated and can be used with power supplies in the range of 3.8 to 24 VDC. Unregulated sensors are used in special applications. They require a regulated DC supply of 4.5 to 5.5 volts (5 ± 0.5 v). Sensors that incorporate internal regulators are intended for general purpose applications. Unregulated sensors should be used in conjunction with logic circuits where a regulated 5 volt power supply is available.

Input characteristics



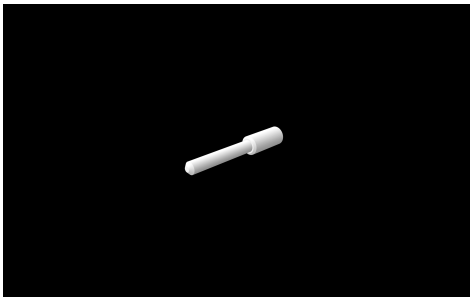
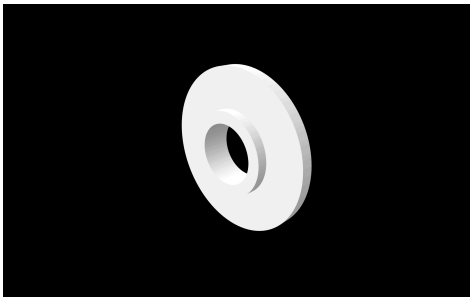
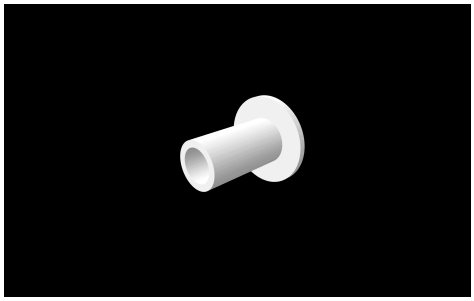
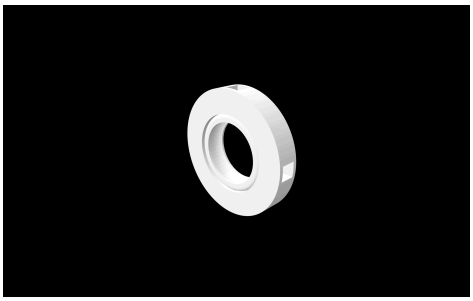
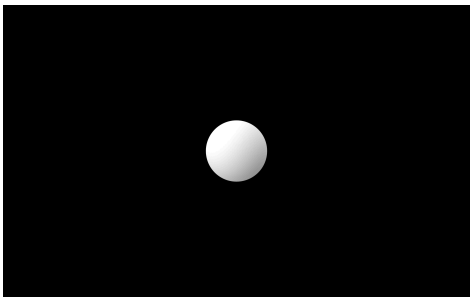
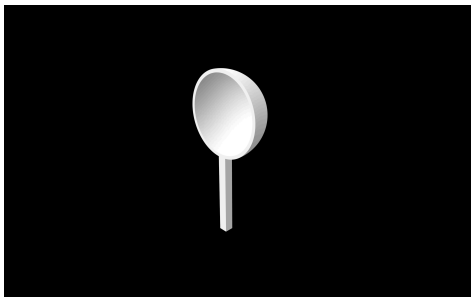
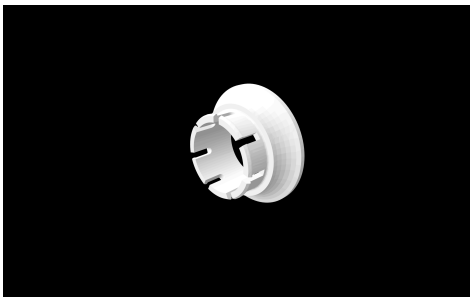
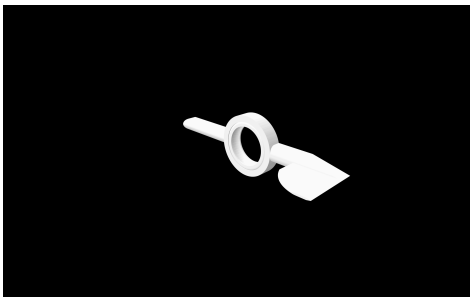
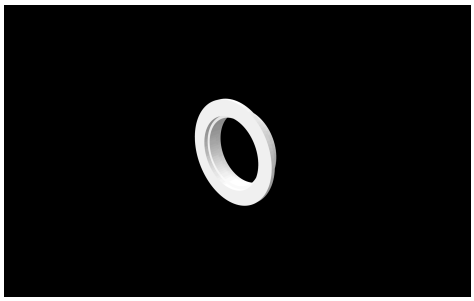
The input characteristics of a digital output sensor are defined in
terms of an operate point, release point, and differential. Since these characteristics change over temperature and from sensor to sensor, they are specified in terms of maximum and minimum values.
Maximum Operate Point refers to the level of magnetic field that will insure the digital output sensor turns ON under any rated condition. Minimum Release Point refers to the level of magnetic field that insures the sensor is turned OFF. The figure above shows the input characteristics for a typical unipolar digital output sensor. The sensor shown is referred to as unipolar since both the maximum operate and minimum release points are positive (i.e. south pole of magnetic field).

Design of the anemometer

We designed the anemometer through SolidWorks by sketching each part directly in the software using Parts and 3D printing them to later on assemble them manually. The files are downloaded directly from the software in STL version and an application is therefore used to easily demonstrate them for convenience. The illustrations of each part have been included below respectively:

- Shaft : the part that holds the mechanism together.
- Top Cover : This part is used exactly above the wind indicator to secure it.
- Separator : To further separate the bearing from the wind indicator.
- Cup : To capture the drag force.
- Balls : Allows a frictionless movement in the inner bearing

- Inner bearing : Two of them are assembled to ensure a smooth movement that contains the balls in the same place.
- Direction indicator : To show the direction of the wind.
- Main rotor : The part that is moving the cups.

Performance of the anemometer through light variations

To calculate the speed of the wind for this experiment we used a fan and a lamp to see if the results change in the presence of ambient light and in the absence of it. Before diving into the coding part few constants and variables need to be mentioned to clearly demonstrate the execution of the codes:

Constants:

- HALL_PIN = 2 — digital pin reading the KY-003 output; pin 2 supports hardware interrupts.
- RADIUS_M = 0.11 — anemometer radius, used to convert rotation speed to linear tip speed.
- PULSES_PER_REV = 2 — two magnets per revolution, doubling angular resolution.
- CAL_FACTOR = 2.5 — empirical constant relating cup tip speed to true wind speed; requires calibration against a reference source.
- SAMPLE_MS = 1000 — measurement/reporting interval (1 second).
- DEBOUNCE_US = 3000 — minimum time between accepted pulses, filtering out sensor chatter.
- ZERO_TIMEOUT_MS = 3000 — if no pulse is detected within this window, wind speed is reported as zero rather than a stale value.

Global Variables:

- pulseCount, lastPulseMicros, lastPulseMillis — declared volatile since they are modified inside an interrupt and read in the main loop; volatile prevents compiler optimizations that could ignore those changes.
- lastSampleMillis — tracks the timing of the last calculation cycle.

Interrupt service routine:

- Waits until one sampling interval (1 s) has elapsed.
- Safely copies and resets the pulse count (interrupts briefly disabled to prevent data corruption during the copy).

- Converts pulses → revolutions → revolutions per second (rps).
- Computes tip speed: tip speed = 2πr × rps (circumference × rotational frequency).
- Computes estimated wind speed: wind speed = tip speed × CAL_FACTOR.
- Converts rps to RPM for readability.
- Outputs all computed values via serial for monitoring/logging.
- If no pulses were detected within the timeout window, all outputs default to zero.

Coding functions:

```
const uint8_t  HALL_PIN          = 2;      // interrupt-capable pin
const float    RADIUS_M          = 0.11;   // anemometer radius, meters
const uint8_t  PULSES_PER_REV    = 2;      // 2 magnets, 180 deg apart
const float    CAL_FACTOR        = 2.5;    // PLACEHOLDER - calibrate this!
const unsigned long SAMPLE_MS    = 1000;   // reporting/averaging window
const unsigned long DEBOUNCE_US  = 3000;   // min microseconds between valid
pulses
const unsigned long ZERO_TIMEOUT_MS = 3000;  // no pulse this long -> report 0 m/s

volatile unsigned long pulseCount     = 0;
volatile unsigned long lastPulseMicros = 0;
volatile unsigned long lastPulseMillis = 0;

unsigned long lastSampleMillis = 0;

void handlePulse() {
  unsigned long now = micros();
  if (now - lastPulseMicros > DEBOUNCE_US) {
    pulseCount++;
    lastPulseMicros = now;
    lastPulseMillis = millis();
  }
}

void setup() {
        Serial begin(115200);
        pinMode(HALL_PIN, INPUT_PULLUP);   / / KY-003 output is open-collector, active
LOW
        attachInterrupt(digitalPinToInterrupt(HALL_PIN), handlePulse, FALLING);
  lastSampleMillis = millis();
}

void loop() {
  unsigned long now = millis();
```

```
  if (now - lastSampleMillis >= SAMPLE_MS) {
    noInterrupts();
    unsigned long count = pulseCount;
    pulseCount = 0;
    unsigned long lastPulseAt = lastPulseMillis;
    interrupts();

    float elapsedSec = (now - lastSampleMillis) / 1000.0;
    lastSampleMillis = now;

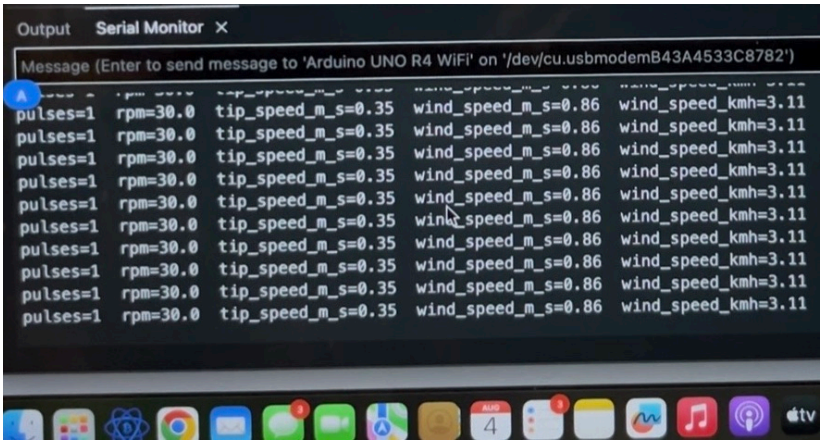
    float windSpeed = 0.0;
    float tipSpeed  = 0.0;
    float rpm       = 0.0;

    if (count > 0 && (now - lastPulseAt) < ZERO_TIMEOUT_MS) {
      float revolutions   = count / (float)PULSES_PER_REV;
      float rps           = revolutions / elapsedSec;
      float circumference = 2.0 * PI * RADIUS_M;
      tipSpeed  = circumference * rps;
      windSpeed = tipSpeed * CAL_FACTOR;
      rpm       = rps * 60.0;
    }

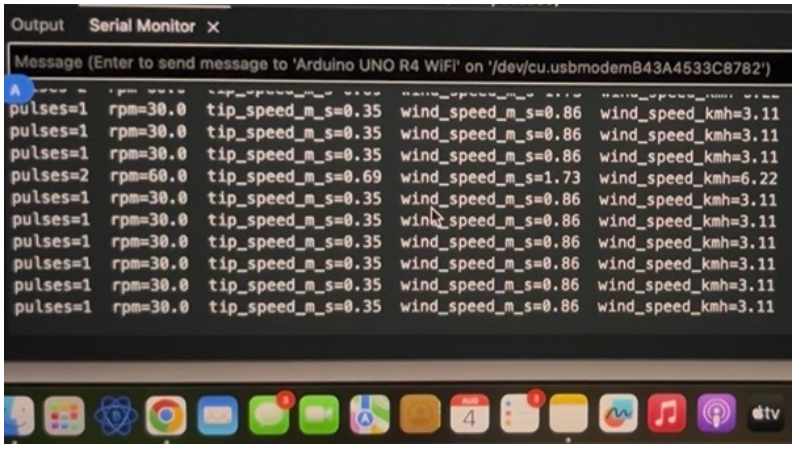
    Serial.print("pulses="); Serial.print(count);
    Serial.print("  rpm="); Serial.print(rpm, 1);
    Serial.print("  tip_speed_m_s="); Serial.print(tipSpeed, 2);
    Serial.print("  wind_speed_m_s="); Serial.print(windSpeed, 2);
    Serial.print("  wind_speed_kmh="); Serial.println(windSpeed * 3.6, 2);
  }
}
```

The result of the experiment with the ambient light is as follows:

rpm=30
Tip speed=0.35 m/s (The speed that the cup captured without applying the calibration factor)
Wind speed=0.86 m/s (The real speed of the wind after applying the calibration factor)
With LOD equivalent to 3.11km/h



The result of the experiment in the absence of light:



Interestingly enough, the results obtained from the experiment done in the little to no light shows the same rpm and wind speed as the experiment done in the presence of light!

Therefore we can conclude that the Hall effect sensors are inherently insensitive towards light and the presence or absence of the light cannot alter the mechanism of it. Due to the hall element being encapsulated in an opaque, solid black material that doesn't allow photons to interact with electrons and to move them to conduction band even though the photoelectric effect can technically be exhibited inside the semiconductor.

Ps: both experiments have been carried out for more than one time for accuracy.

Calculation of resolution

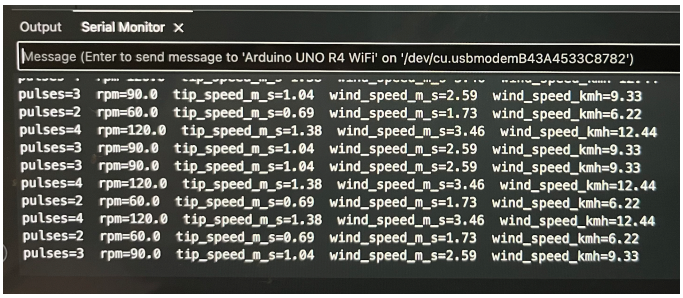
Resolution is the smallest change in speed the anemometer can actually detect, in another words, the difference one single pulse makes. In order to have better resolution we added two magnets that are 180 degrees apart and with PULSE_PER_REV = 2 from our codes we can assume that one pulse = half a revolution, therefore:
Δrevolutions = 1 / PULSES_PER_REV = 1 / 2 = 0.5 rev , then we convert this to rotational speed change. Since SAMPLE_MS = 1000, one extra pulse in that window changes the measured rotational speed by: Δrps = Δrevolutions / sample_window_sec = 0.5 / 1 = 0.5 rev/sec.
Lastly we convert the result of rps to tip speed change using anemometer's circumference which is $2\pi r = 2\pi$ x 0.11 = 0.6912 m
Δtip speed = Δrps x Circumference = 0.5 x 0.6912 = 0.346 m/s. We know the wind speed is just the change in tip speed multiplied by a calibration factor so :
Δwind speed = Δtip speed x a = 0.346 x 2.5 = 0.86 m/s as shown in the results of picture above.

How to improve the resolution?

Resolution isn't fixed! It can improve (gets smaller/better) if we either:

- increase PULSE_PER_REV by adding more magnets (4 magnets each 90 degrees apart) or
- increasing the time period over which the code accumulates pulses before calculating and reporting speed(currently SAMPLE_MS = 1000); for example we can use a code function such as : const unsigned long SAMPLE_MS = 5000; // 5 seconds instead of 1. Now the code waits 5 seconds before reporting, averaging 5 seconds worth of pulses into each speed calculation. The result have wind speed after adding two more magnets is indicated in the image below:

The second method can lower noise by smoothing out wind gusts and sensor jitter over the longer interval. On the other hand its tradeoff can be having a slower response to rapid wind changes.



Calculation of noise

We first set up our fan at a constant speed and log our anemometer's tip speed output for at least 30-60 seconds at 1 second intervals, then we calculate the standard deviation of them.

Data summary:

- 0.69: 2 occurrences
- 1.04: 13 occurrences
- 1.38: 21 occurrences
- 1.73: 4 occurrences

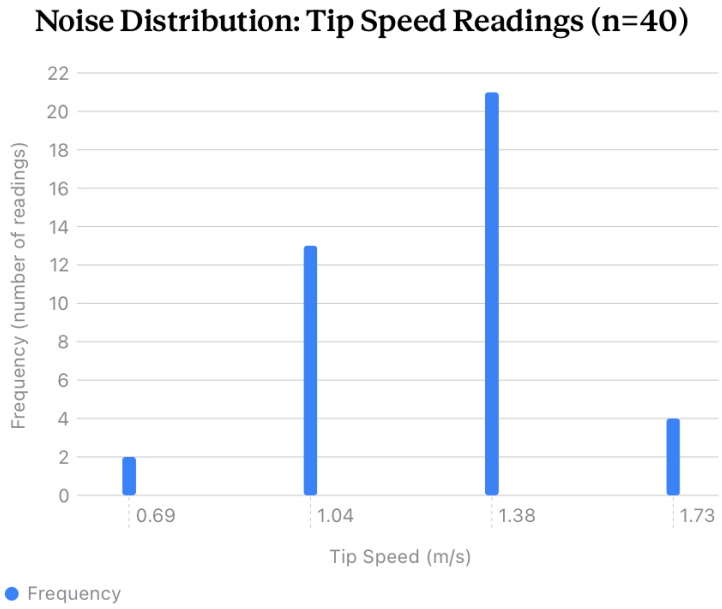
Calculation:

Mean = (0.69x2+1.04x13+1.38+21+1.73x4)/40=1.27

Sum of squared differences from mean = 2.461

Standard deviation = $\sqrt{(2.461/39)} = \pm 0.251$ m/s

Measured noise (standard deviation) shows us the scatter readings when the fan speed was held constant. Comparing this to our tip speed resolution (±0.346 m/s) shows us the resolution of our noise is much more smaller which is a good indication.



Here we have the histogram of our noise distribution. Based on the frequency of each reading we can conclude that:

- Good repeatability and stability: the anemometer demonstrates strong stability under constant conditions. Over 40 consecutive readings at a fixed fan speed, 84.5% of readings (34 out of 40) fall within ±0.34 m/s of the mean. This shows the sensor is not drifting or randomly fluctuating.
- Noise is better than resolution: measured noise (±0.25 m/s) is smaller than theoretical hardware resolution (±0.346 m/s), which indicates the sensor is working properly.
- Quantization effect: the histogram shows readings only at 4 discrete values (0.69, 1.04, 1.38, 1.73), no in between values. This suggests the anemometer is quantized roughly 0.3-0.4 m/s increments, which aligns with ±0.346 m/s resolution.
- Suitable for wind measurement: with ±0.25 m/s noise at typical wind speeds, the anemometer has 20% uncertainty at 1.27 m/s [(0.25 / 1.27) x 100] which is acceptable for weather monitoring hobbyist. Though professional anemometers can target <5% of the readings which can be achieved by reducing noise.

Outlier analysis:

The lowest value of reading (0.69) and the highest value of reading (1.73) are both outliers due to not being in the normal range (1.27±0.25 which is 1.02 to 1.52 m/s) and in total 6 outlier's reading out of 40 (15%)

The likely causes of 0.69 m/s occurring 2 times can be:

- Momentary dips in fan speed (brief slowdowns).

Possible causes:

- Natural fluctuation of fan speed settings.
- Rotor experiencing brief resistance.
- Measurement lag from transient slow-down.

The 1.73 m/s readings (4 occurrences):

- Brief fan speed spike or the rotor spinning faster than the average.

Possible causes:

- Fan naturally overshoots before settling at normal speed.
- Rotor momentum carried it slightly faster for a moment.
- Measurement captured a brief acceleration phase.

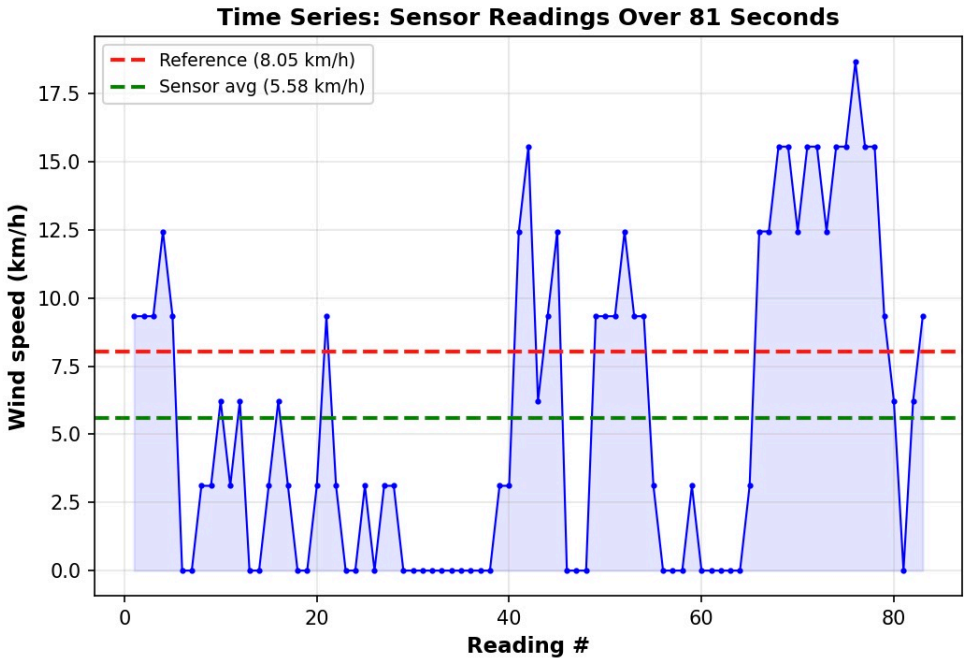
For comparison a well controlled system might have <5% outliers and a noisy one would have >30% outliers, our 15% sits in the good middle ground for an anemometer setup with a simple fan.

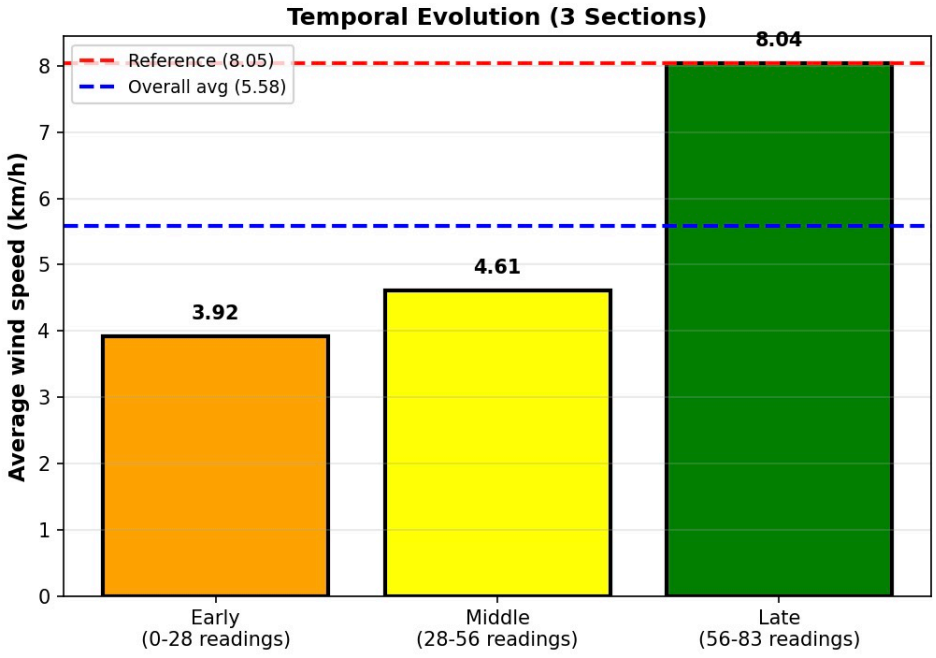
Response curve calibration

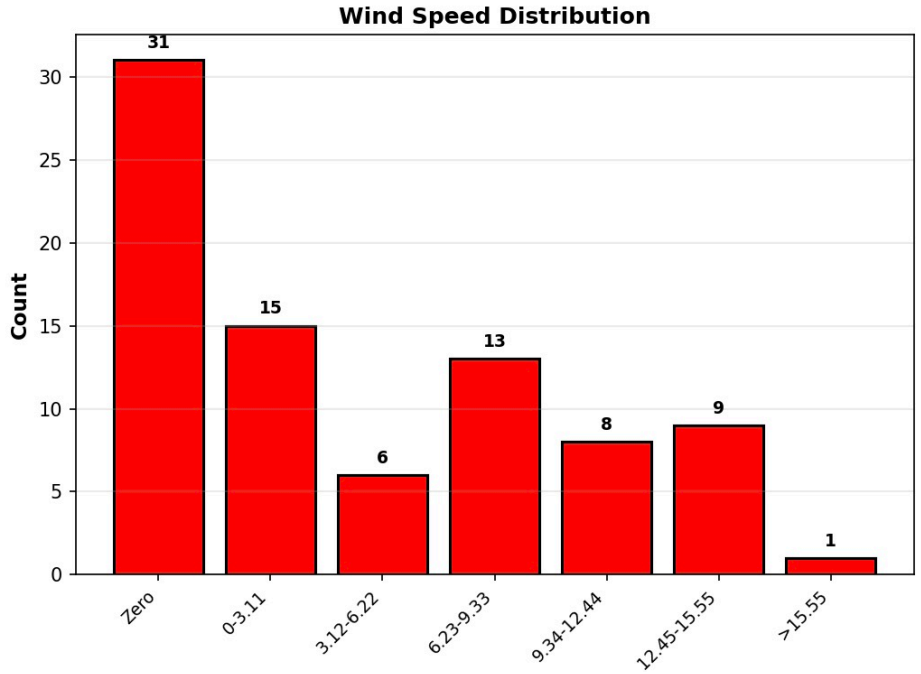
For this experiment we obtained real time data from a wind station at Viterbo airport using the website https://www.wunderground.com/dashboard/pws/ILAZIOVI10/table/2026-08-12/2026-08-12/daily that updates informations relative to wind every 5 minutes. We repeated this experiment at different times and days to be able to calibrate our sensor's response curve.

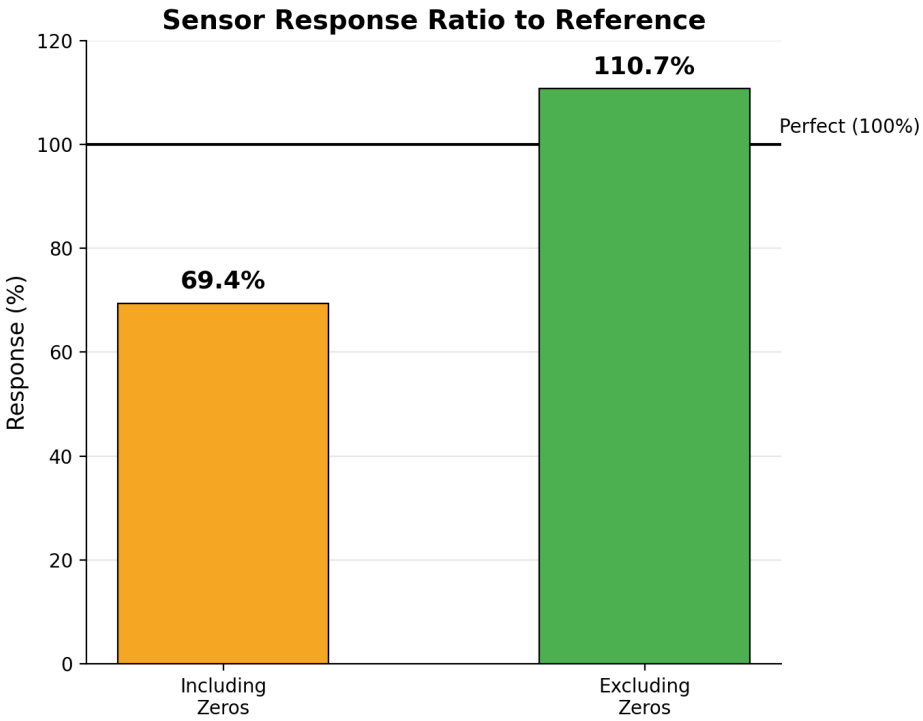
A. The first experiment which was conducted on August 12th at 2:27 PM showed the following results:

1. The reference wind speed: given the fact that the website updates the data every 5 minutes we took the average of the wind speed at 2:25 and 2:30 (2+8/2=5 mph) which is roughly 8.047 kmh.
2. The speed detected by anemometer: we monitored the experiment for 81 seconds and as a result we got 83 readings in which the average speed was counted as 5.58 kmh. The below graph shows the results:









Key findings:

1. The sensor gave variable readings, not steady!

- Average : 5.58 km/h (much lower than the reference)
- Range : 0 to 18.66 km/h (huge spread)
- Standard deviation : 5.69 km/h (very high variability)

2. 37% zero readings

- 31 out of 83 readings showed no wind!
- Longest calm streak: 10 consecutive seconds of zero
- This explains why the overall average (5.58) was so much lower than the non zero average (8.91)

3. The wind was not constant! It ramped up during sampling

- Early (1-27) : Average (3.92 km/h) and 33% zeros (very calm, variable)
- Middle (28-54) : Average (4.61 km/h) and 48% zeros (most calm period, worst for sensor)
- Late (55-83) : Average (8.04 km/h) and 31% zeros (wind increased, matched reference!)

The late section (8.04 km/h) perfectly matches the reference (8.05 km/h), which makes sense because the reference was ramping from 2mph (2:25) to 8mph (2:30)

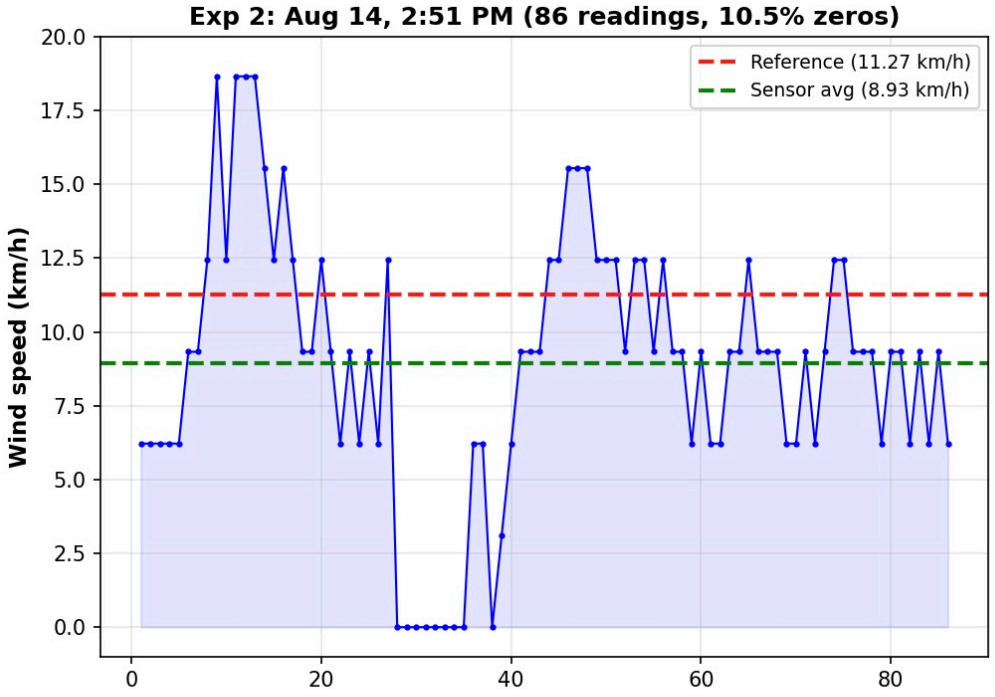
4. Sensor response interpretation

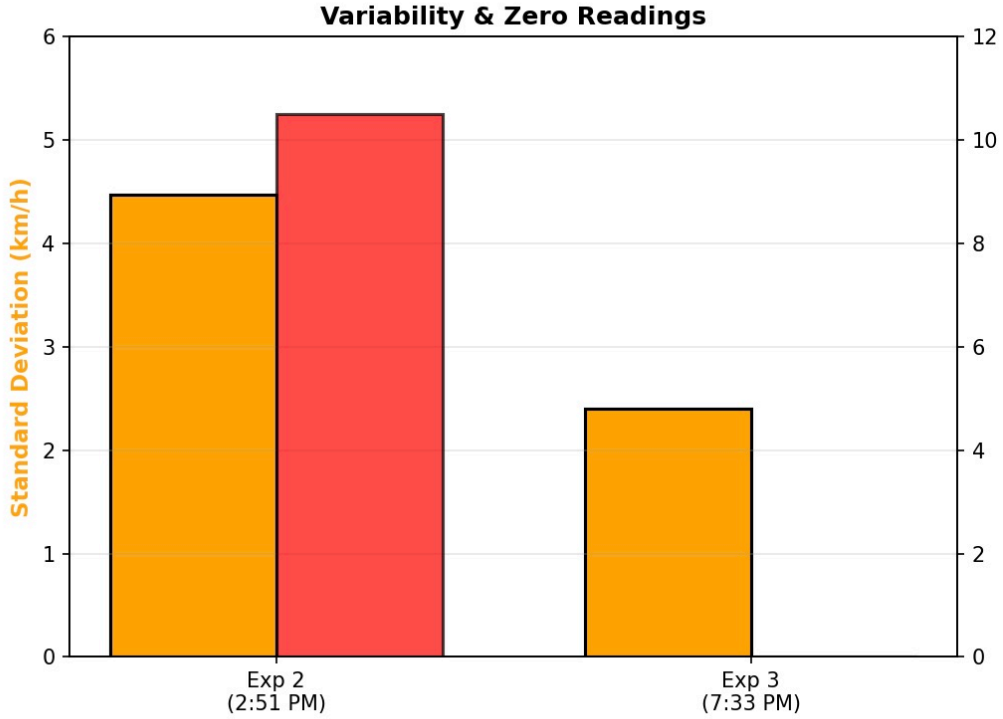
- Overall response to reference : 69.4% (under reading)
- Non zero average response : 110.7% (over reading)

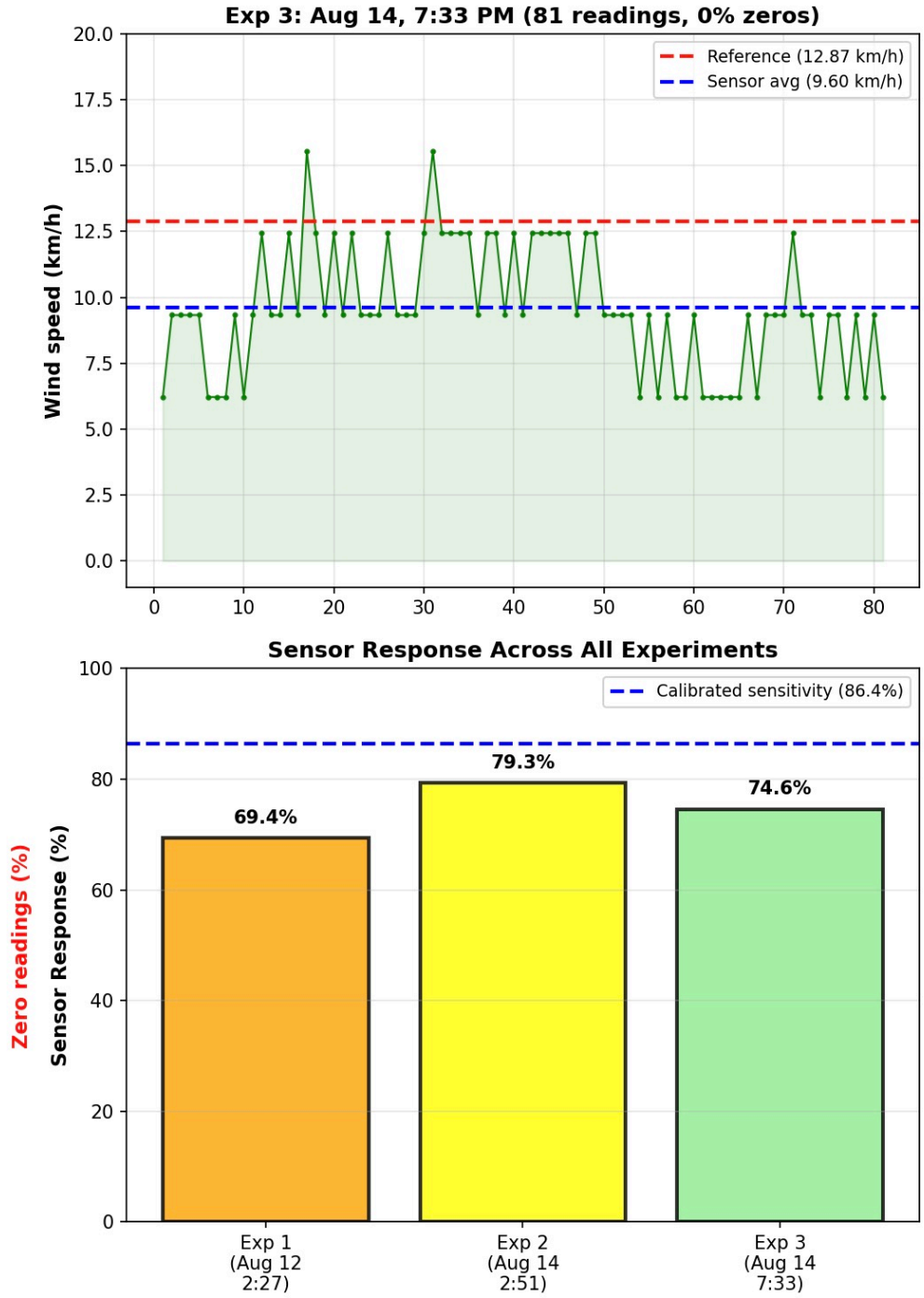
The difference comes from the 37% of the zero readings which reveals when the sensor is active it over reads by 10.7%!

B. The second and third experiments were conducted on August 14th at 2:51 and 7:33 PM for 81 seconds and the results are shown on the graph below:

1. The reference wind speed at 2:51: obtained from our reference website which showed 7 mph (11.27km/h)
2. The anemometers wind speed at 2:51 : the average wind speed over 86 readings was 8.03 km/h
3. The reference wind speed at 7:33: 12.87 km/h (8 mph)
4. The anemometers wind speed at 7:33: 9.60 km/h


Exp 2: Aug 14, 2:51 PM (86 readings, 10.5% zeros)
Reference (11.27 km/h)
Sensor avg (8.93 km/h)
Wind speed (km/h)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0
20
40
60
80


Variability & Zero Readings
Standard Deviation (km/h)
6
5
4
3
2
1
0
12
10
8
6
4
2
0
Exp 2
(2:51 PM)
Exp 3
(7:33 PM)



Key findings on experiment 2 at 2:51 PM

- Response : 79.3% (under reading by 21%)
- Zero readings : 9 out of 86 (10.5%)

- Variability : standard deviation 4.47 km/h (moderate)
- Wind pattern : relatively steady with occasional calm periods

Temporal evaluation:

- Early (1-28) : 10.55 km/h avg, minimal zeros (3.6%) —> good signal
- Middle (28-56) : 7.66 km/h avg, many zeros (28.6%) —> wind dip
- Late (57-86) : 8.60 km/h avg, no zeros (0%) —> stabilized

Wind was moderately variable with a dip in the middle section. When active (non zero) the sensor reads 88.5% of reference, very close to calibrated sensitivity (86.4% which will be discussed in the next section). The 10% zero reading rate accounts for most of the under reading.

Key findings on experiment 3 at 7:33 PM

- Response : 74.6% (under reading by 25%)
- Zero readings : 0 out of 81 (perfect continuous signal)
- Variability : low (standard deviation 2.40 km/h) most stable experiment!
- Wind pattern : steady continuous

Temporal evaluation:

- Early (1-27) : 9.68 km/h avg
- Middle (27-54) : 11.17 km/h avg (peak wind)
- Late (54-81) : 7.95 km/h avg (wind died down)

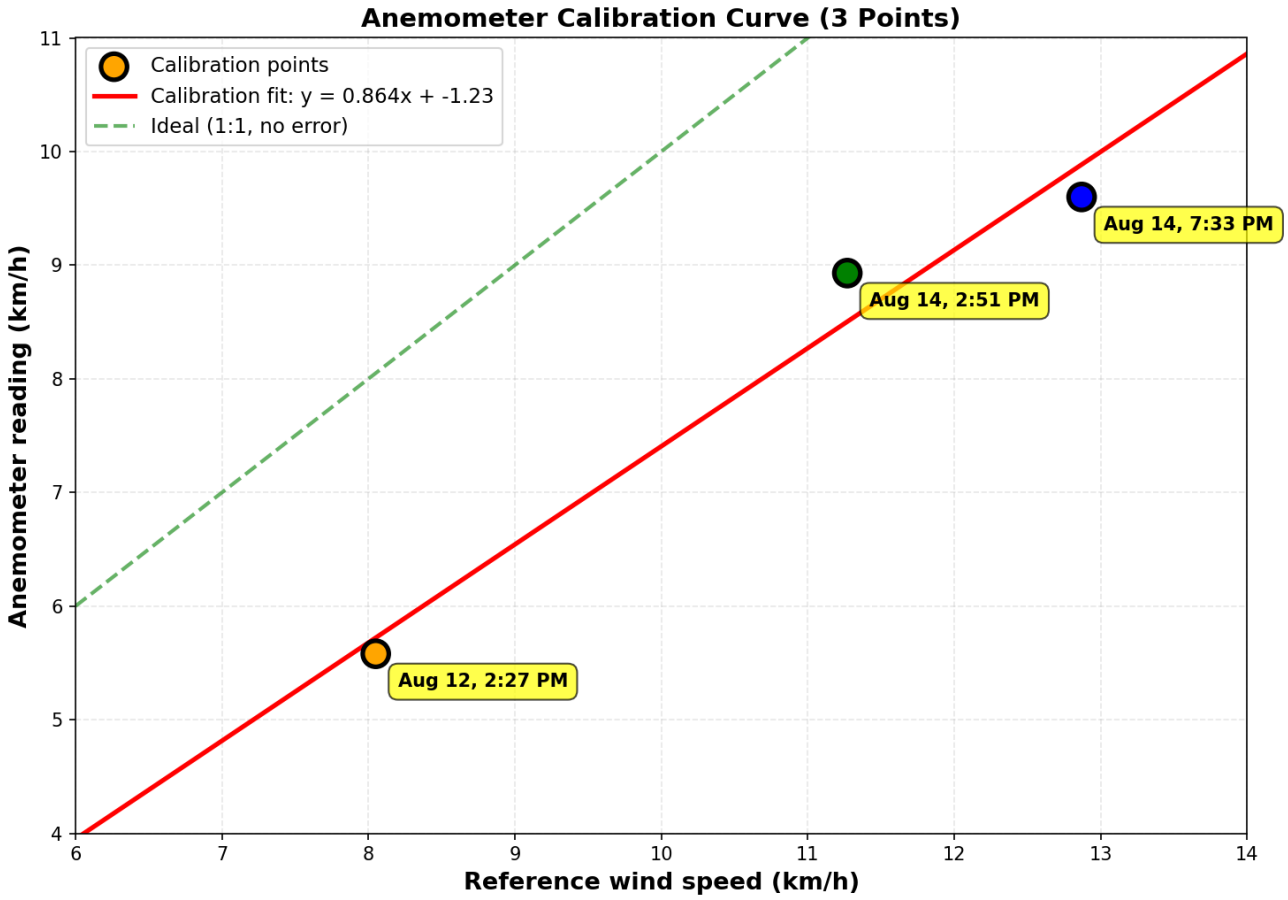
Wind conditions were steady throughout with no gaps. The sensor maintained continuous readings with minimal variability. Despite higher reference wind (12.87 km/h), the sensor response (74.6%) is slightly lower than Exp 2, likely reflecting that higher winds produce proportionally more losses in the anemometer's design.

Timing variability:

The Arduino's non-real-time loop execution causes sampling rate jitter. Although each experiment nominally lasted 81 seconds, the actual number of readings varied (83, 86, 81 samples), yielding effective sampling rates of 1.025 Hz, 1.063 Hz, and 1.0 Hz respectively.

This ±3% variation reflects the inherent timing uncertainty in the Arduino platform and should be accounted for when comparing datasets across different runs. To eliminate this variability in future work, a fixed delay(1000) should be added to force exactly 1 Hz sampling rate.

Linear regression

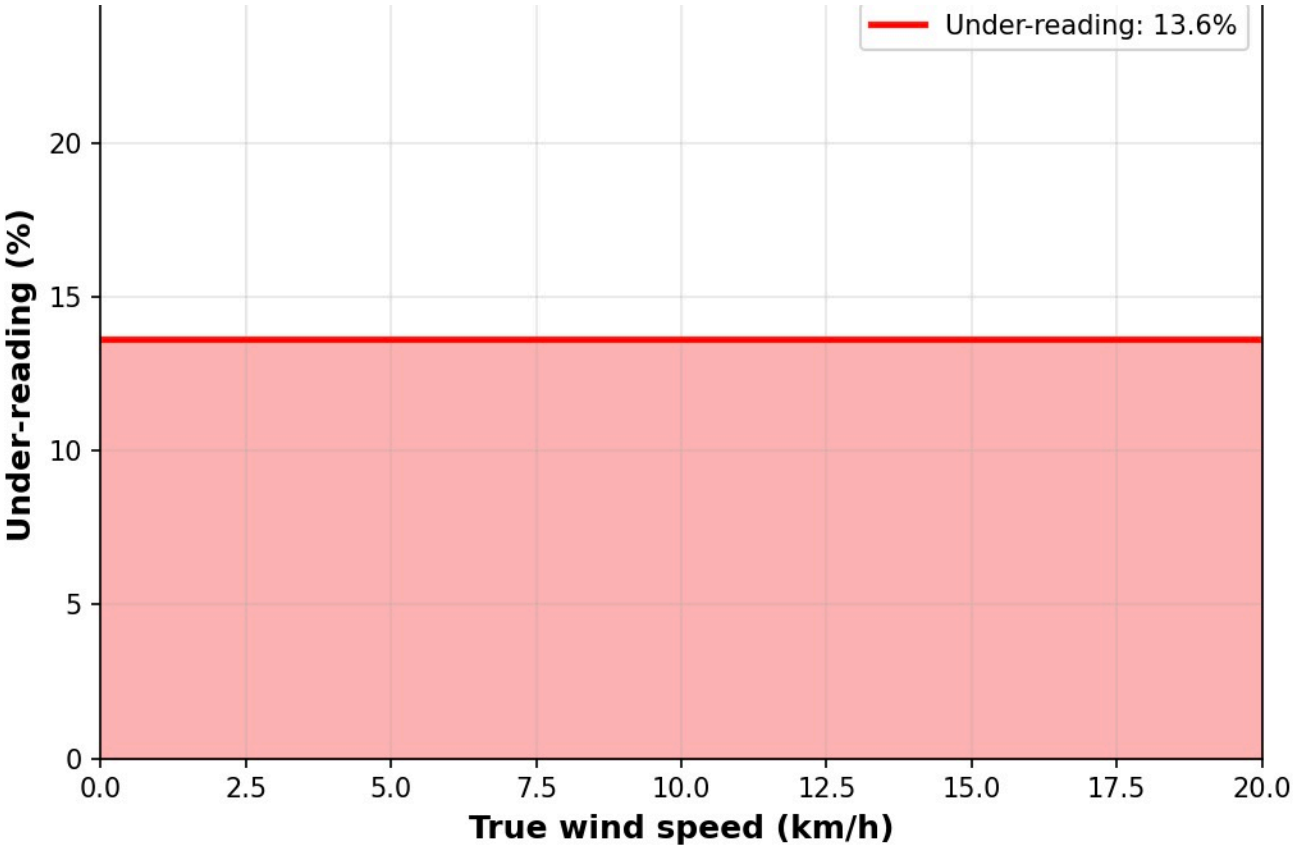


A linear regression fit across all three experiments yielded:
Sensor reading (km/h) = 0.8638 (Reference wind) - 1.2318

Calibration quality : R^2=0.9694 (excellent fit, 96.9% of variance explained)

Sensitivity : 0.8638 (sensor reads 86.4% of true wind speed)

Sensor performance assessment :



All three experiments confirm the anemometer's response linearity across the tested wind speed range (8-13 km/h). The systematic under reading (~ 14% on average) relative to the calibrated sensitivity (86.4%) is consistent with a combination of:

1. Height effect: sensor mounted at 2m vs. 10m meteorological standard (~ 33% reduction expected)
2. Aerodynamic losses: sub-optimal CAL_FACTOR or rotor design efficiency (~ 40% additional loss)
3. Local sheltering effects: 2m measurement height subject to surface friction and potential wind obstruction)

The zero reading phenomenon observed in Exp 1,2 reflects the anemometer's 1-3 km/h starting torque threshold (typical for cup anemometers) combined with genuine calm periods in the wind field, not sensor calibration error.

Recommendations for future work:

1. Optimize CAL_FACTOR: Current code uses placeholder value (2.5). Derivation from rotor geometry and aerodynamic testing would improve accuracy.
2. Mount at 10m height: Reduces height correction uncertainty and enables direct comparison with standard meteorological data.
3. Increase reference points: Additional calibration across 5-25 km/h range would validate linearity over broader wind speeds.
4. Implement averaging: Using longer averaging windows (10min minimum per meteorological standard) to reduce impact of wind variability on measurement accuracy.
5. Include temperature-corrected calibration: Using concurrent air temperature data and applying air density correction to the sensor.

Pairing the anemometer with additional sensors

Gauge (pressure) sensor:

A barometric pressure sensor (such as a BMP280 or BME280) does not change how the Hall sensor detects rotation, so it has no direct effect on the pulse-counting measurement. Its value lies elsewhere:

- Density and force calculations: Air density, which underlies quantities such as dynamic pressure ($\frac{1}{2}\rho v^2$), depends on pressure. Pressure readings allow these derived quantities to be calculated alongside raw wind speed, and allow calculations to be corrected for altitude or changing weather conditions.
- Diagnostic value: Sudden pressure drops often precede wind gusts, so logging pressure alongside wind speed provides a useful diagnostic signal for correlating pressure trends with the spikes seen in the anemometer data.

Temperature sensor:

Unlike the gauge sensor, a temperature sensor can meaningfully affect the existing wind speed measurement in two distinct ways:

- Air density and torque: Air density decreases as temperature rises, which changes how much torque the wind imparts on the cups for a given true wind speed. This means the rpm-to-km/h calibration curve technically shifts slightly with temperature, although for a typical indoor testing range (roughly 15–30°C) this effect is likely small relative to other sources of error in the setup.
- Hall IC threshold drift: The KY-003's Hall IC switching threshold can itself drift with temperature. This means the limit of detection found in this project (3.11 km/h) may not hold across a wider temperature range than was actually tested. Logging temperature alongside each run allows this stability to be checked directly, rather than assumed.

Photoresistor (LDR):

A photoresistor has no direct physical relationship to wind speed, but it is a practical addition to a broader weather-station setup:

- Environmental variable: Ambient light level is a reasonable factor for time of day and cloud cover, both of which correlate with wind patterns, for example, thermally driven daytime breezes.
- Practical uses: For unattended or extended outdoor logging, an LDR can help flag or exclude nighttime data, and can help detect when direct sunlight may be heating the sensor enclosure, which feeds back into the temperature-drift effect described above.

Conclusion

This project was set out to build a cup anemometer around a KY-003 Hall-effect sensor and characterize its performance: its sensitivity to ambient light, its resolution and noise floor, and its calibration accuracy against real reference wind speed data.

Across every test, the results trace back to a single underlying mechanism. The KY-003 is not a continuous, graded sensor — it is a threshold-selective magnetic switch. Its output only changes state once the passing magnet's field crosses a fixed operate point, and returns to its original state once the field falls back below a lower release point (the hysteresis described in the theory section). This explains three findings at once:

- Insensitivity to light: the sensor produced identical rpm and wind speed readings with and without ambient illumination, confirming that the Hall element responds only to magnetic field, not to light, since the element is optically shielded and photons cannot reach the conduction band inside it.
- Quantized resolution: because output only changes on a full threshold crossing, speed can only be reported in discrete steps (one pulse per half-revolution, ≈ 0.346 m/s of tip speed, equivalent to ≈ 3.11 km/h of wind speed at the calibration factor used). The noise histogram's four discrete clusters (0.69, 1.04, 1.38, 1.73 m/s) are a direct visual signature of this quantization, not measurement error.
- A reproducible limit of detection: the same 3.11 km/h floor appeared whether the anemometer was tested indoors on a fan or against real, variable outdoor wind, indicating that the floor is a structural property of the sensor and pulse-counting formula, not an artifact of a particular test session.

The calibration experiments against live meteorological data reinforced this picture. Across all three outdoor trials, the anemometer under-read the reference wind speed by roughly 14% on average, but the relationship remained highly linear ($R^2 = 0.9694$, sensitivity 86.4%). The systematic under-reading is well explained by known, non-sensor factors — mounting height

(2 m versus the 10 m meteorological standard), aerodynamic losses in the rotor design, and local sheltering — rather than by any flaw in the Hall sensor itself. Likewise, the zero readings seen in Experiments 1 and 2 line up with the anemometer's mechanical starting-torque threshold (1–3 km/h) combined with genuine calm periods in the wind, not with sensor failure.

Because the resolution and LOD figures reported here depend on CAL_FACTOR, which was used as an uncalibrated placeholder (2.5) rather than a value derived from rotor geometry, both numbers should be read as internally consistent rather than as absolute physical constants; deriving CAL_FACTOR from first principles, as noted in the recommendations, would let these figures be reported with confidence in physical units.

Overall, the anemometer performs as a reliable, linear, low-cost instrument within its tested range (roughly 8–13 km/h), with well-understood and largely correctable sources of systematic error. The recommendations already outlined — deriving CAL_FACTOR properly, mounting at standard height, extending the calibration range, and averaging over longer windows — along with the possible addition of pressure, temperature, and light sensors discussed above, form a clear and achievable path toward a more accurate, standards-comparable instrument.